Time = t = OE = DA

From the graph we know that, AB = DC

First equation of motion

By definition, Acceleration

= Change in velocity / Time

= (Final velocity – Initial velocity)/Time

= (OC – OD) / OE

= DC / OE

a = DC / t

DC = AB = at

From the graph EB = EA + AB

$$v = u + at \quad (1)$$

This is the first equation of motion.

Second equation of motion

From the graph the distance covered by the object during time, t is given by the area of quadrangle DOEB

s = Area of the quadrangle DOEB

= Area of the rectangle DOEA + Area of the triangle DAB

= (AE × OE) + (1/2 × AB × DA)

$$s = ut + \tfrac{1}{2}at^2 \quad (2)$$

This is the second equation of motion.

Third equation of motion

We see that the distance covered by the object during time, t is given by the area of the quadrangle DOEB. Here, DOEB is a trapezium. Then,

s = Area of trapezium DOEB

= ½ × Sum of length of parallel side × Distance between parallel sides

= ½ × (OD + BE) × OE

s = ½ × (u + v) × t

Since, a= (v – u) / t or t = (v – u)/a

s = ½ × (v + u) × (v – u)/a

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as \quad (3)$$

This is the third equation of motion.

Problem 3

The brakes applied to a car produce an acceleration of 6 ms^{-2} in the opposite direction to the motion. If the car takes 2 s to stop after the application of brakes, calculate the distance traveled during this time.

Solution:

We have been given a = $-6\ ms^{-2}$, t = 2s and v = 0

From the equation of motion,

v = u + at

0 = u + (−6 × 2)

0 = u – 12 $\quad \therefore u = 12\ ms^{-1}$

$s = ut + \tfrac{1}{2}at^2$

= (12 × 2) + ½ (−6 × 2 × 2)

= 24 – 12 = 12m

Thus, the car will move 12 m before it stops after the application of brakes.

2.7 Motion of freely falling body

Activity 4

Take a large stone and a small eraser. Stand on the top of a table and drop them simultaneously from the same height. What do you observe? Now, take a small eraser and a sheet of paper. Drop them simultaneously from the same height. What do you observe? This time, take two sheets of paper having same mass and crumple one of the sheets into a ball. Now, drop the sheet and the ball from the same height. What do you observe?

From Activity 4, you can observe that, both the stone and the eraser reach the surface of the earth almost at the same time. When you drop the eraser and paper, the eraser reaches first and the sheet of paper reaches later. You can also observe that the paper crumpled into a ball reaches ground first and plain sheet of paper reaches later, although they have equal mass. Do you know the

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